积分表

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a}+C$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a}+C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \ln|x+\sqrt{x^2-a^2}|+C$$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln(x+\sqrt{x^2\mp a^2})+C$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|+C$$

$$\int \tan x\,dx = -\ln|\cos x|+C$$

$$\int \cot x\,dx = \ln|\sin x|+C$$

$$\int \sec x\,dx = \ln|\sec x+\tan x|+C$$

$$\int \csc x\,dx = \ln|\csc x-\cot x|+C$$

$$\int \frac{1}{y\ln y}\,dy = -\ln\frac{1}{\ln y}$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\arccos x)' = \frac{-1}{\sqrt{1-x^2}}$$

$$(\arctan x)' = \frac{1}{1+x^2}$$

$$(\operatorname{arccot} x)' = \frac{-1}{1+x^2}$$

$$(x\ln x - x)' = \ln x$$

$x^3+x^2+x+1 = (x^2+1)(x+1)$

$x^3-x^2+x-1 = (x^2+1)(x-1)$

$x^3+x^2-x-1 = (x^2-1)(x+1)$

$x^3-x^2-x+1 = (x^2-1)(x-1)$

万能替换法

$x = \tan\frac{u}{2}$

$dx = \frac{1}{\frac{1}{2}\left(1+\left(\frac{u}{2}\right)^2\right)}\,du$

$\frac{1}{\tan x} \to \cot x$　　$\frac{1}{\sin x} \to \csc x$　　$\frac{1}{\cos x} \to \sec x$

分数积分

(1)

(2) 真分数

① 拆项设 A、B、C、D 解方程

② 观察分子分母关系：若 (分母)' = 分子，则 $\ln|\text{分母}|$

求极限中 ① 上下同除最高次数

② 洛必达法则 ($\frac{0}{0}$ 或 $\frac{\infty}{\infty}$)

(3) 假分数

化成 常数 + 真分数

分部积分

形如 $\arctan x$　　xe^x

每日默写

一些三角恒等式.

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a}+C$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a}+C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \ln|x+\sqrt{x^2-a^2}|+C$$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln(x+\sqrt{x^2+a^2})+C$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\frac{x-a}{x+a}.$$

$$\int \tan x\,dx = -\ln\cos x + C$$

$$\int \cot x\,dx = \ln\sin x + C$$

$$\int \sec x\,dx = \int \frac{1}{\cos x}dx = \ln|\sec x+\tan x|+C$$

$$\int \csc x\,dx = \int \frac{1}{\sin x}dx = \ln|\csc x-\cot x|+C$$

$$\int \frac{1}{y\ln y}dx = -\ln\ln y$$

$$\tan(\alpha\pm\beta) = \frac{\tan\alpha\pm\tan\beta}{1\mp\tan\alpha\tan\beta}$$

和差化积用和差角公式推